Before the
National Science Foundation
Alexandria, VA
and the
Office of Science and Technology Policy
Washington, DC

In re REQUEST FOR INFORMATION ON THE DEVELOPMENT OF AN ARTIFICIAL INTELLIGENCE (AI) ACTION PLAN	F.R. Doc. 2025-02305

COMMENTS OF
COMPUTER & COMMUNICATIONS INDUSTRY ASSOCIATION

The Computer & Communications Industry Association (CCIA)[1] submits the following comments in response to the National Science Foundation's February 6, 2025, Request for Comments on behalf of the Office of Science and Technology Policy.[2]

CCIA is an international, not-for-profit trade association representing a broad cross section of communications and technology firms. For more than 50 years, CCIA has promoted open markets, open systems, and open networks. CCIA members employ more than 1.6 million workers, invest more than $100 billion in research and development, and contribute trillions of dollars in productivity to the global economy.

CCIA members are at the forefront of research and development in technological fields such as artificial intelligence (AI) and machine learning, semiconductor manufacturing, and other computer-related inventions. Many of the key innovations in modern artificial intelligence have stemmed from research done by CCIA members, including the original development of transformer models by Google and the development and open-sourcing of frameworks for deep learning such as Meta's PyTorch and Google's TensorFlow. A regulatory environment that enables continued development of these innovations is essential to the continued development of artificial intelligence in the United States.

I. Summary

Policy at the federal level is critical to ensuring that the U.S. continues to lead the world in AI technology. In order to facilitate U.S. leadership, it is critical to take action in a number of arenas ranging from energy policy to export controls to privacy. Key steps that the federal government can take include:

[1] A list of CCIA members is available online at https://www.ccianet.org/about/members.
[2] Request for Information on the Development of an Artificial Intelligence (AI) Action Plan, 90 Fed. Reg. 9088 (Feb. 6, 2025).

- Ensuring access to sufficient AI infrastructure, including data centers, energy, skilled workforce, and hardware.
- Adopting balanced export control and trade policies that ensure national security without overly restricting access to foreign markets.
- Creating a unified approach to AI policy nationally employing a risk-based approach.
- Assuring data access, including by strongly supporting fair use in the U.S. and similar text and data mining exceptions around the world.
- Avoiding the abuse of patents to restrict AI development.
- Preserving a role for open source AI models in the ecosystem.
- Protecting the privacy of Americans without stifling AI development.

These basic actions, described in more detail below, would help to ensure that the U.S. continues to lead the world in the development and use of artificial intelligence.

II. Developing AI Infrastructure

To foster advancements in artificial intelligence (AI) within the United States, it is essential to implement federal policies that provide the needed access to infrastructure and workforce development. These measures will create an ecosystem conducive to innovation, ensuring that AI technologies can thrive while maintaining security and fairness.

A. Energy access

The growth of AI heavily depends on reliable energy supplies, particularly for data centers that require substantial electricity supply to power advanced computing infrastructure.[3] Enhancing federal authority over transmission infrastructure will help to reliably support the efficient operation of AI technologies. By granting greater power to agencies to site and permit interstate transmission lines, the U.S. would ensure a reliable energy supply for critical AI infrastructure such as data centers. Streamlining approval processes and, where necessary, utilizing eminent domain can facilitate faster deployment of these critical facilities. A potential home for this authority would be the Federal Energy Regulatory Commission (FERC).[4]

It is also crucial that the federal government prevent states from hindering access to the necessary inputs for data centers. To maintain a uniform legal regime across the country, federal policies must prevent states from enacting inconsistent or discriminatory measures that impose regulatory burdens on data centers. Similarly, ensuring equal access to economic incentives will safeguard against undue compliance costs and regulatory barriers to entry, fostering a competitive environment where AI technologies can flourish without unnecessary obstacles.

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[3] *See, e.g.*, EPRI, *Powering Intelligence: Analyzing Artificial Intelligence and Data Center Energy Consumption* (May 28, 2024), https://www.epri.com/research/products/000000003002028905.
[4] *See, e.g.*, Jay Obernolte, *Comments of Congressman Jay Obernolte Re: Docket Nos. EL25-20, ER24-2172, and ER24-2888 et al* (Dec. 5, 2024), https://elibrary.ferc.gov/eLibrary/docinfo?accession_number=20241209-4000.

Finally, investment in domestic nuclear power is likely to prove essential to energy access for data centers.[5] Domestic nuclear power, ranging from traditional large-scale power plants to innovative new technologies such as small modular reactors, offers a reliable and low-carbon energy source vital for data centers. By providing streamlined regulatory processes, tax incentives, and loan guarantees, the Administration could encourage the development of this critical energy option, supporting both environmental goals and AI infrastructure needs. Further, streamlined and consistent regulatory frameworks also make it easier for companies to comply with regulatory requirements, thereby helping to achieve the intended regulatory outcomes.

B. Workforce development

Energy alone cannot operate a data center or develop AI. A skilled workforce is fundamental to sustaining AI advancements. While educational options for AI development are sufficiently well-supported, with computer science programs across the country offering courses of study in artificial intelligence technology, the same is not necessarily true for necessary supporting roles such as manufacturing and data center technicians. In order to maintain American leadership in AI, we must also invest in world-class training programs for these critical roles.[6]

To provide the necessary pool of trained technicians, a number of strategies are available. Partnerships between educators and industry can help to align curricula with real-world demands, ensuring that students gain relevant skills and providing a pathway to a career. In parallel, investing in community colleges and vocational schools will expand access to training programs for individuals from a variety of backgrounds, building a broader workforce and providing benefits across the entirety of the country. Providing additional financial assistance targeted at these career paths will also incentivize entry into these jobs while reducing the burden on those who come out of these programs.

Additionally, support for retraining initiatives can help to mitigate any disruptions to the labor market stemming from increasing use of artificial intelligence. As AI adoption grows, some jobs may be automated. But at the same time, new opportunities will emerge in fields such as data center operations, AI system maintenance, and cybersecurity. By offering retraining programs tailored to these roles, we can ensure that Americans are not left behind by automation but instead can move into new roles. This approach both supports workforce development and helps those whose lives might be disrupted by the growth of these new technologies.

C. Security considerations

When establishing security requirements for international data centers, it is important to maintain the FedRAMP moderate level as generally sufficient. This approach balances security with operational efficiency, allowing businesses to adhere to standards without unnecessary complexity when operating abroad. While it may be appropriate to designate specific countries which require additional security standards due to local conditions, such as geopolitical instability or heightened cyber-crime threats, any such designation should be clear and specific to avoid unnecessary compliance burdens.

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[5] *See, e.g.*, FTI Consulting, *The Powerful Duo of Nuclear and Data Centers* (Feb. 27, 2025), https://www.fticonsulting.com/insights/articles/powerful-duo-nuclear-data-centers.
[6] *See, e.g.*, IT Modernization Centers of Excellence, *AI Guide for Government* Ch. 4 (2025), https://coe.gsa.gov/coe/ai-guide-for-government/developing-ai-workforce/index.html.

In addition, new background-check requirements for cloud service provider (CSP) employees should be designed to avoid undue burdens and costs while respecting existing multi-layered technical controls. This balanced approach ensures data security without deterring skilled workers, maintaining both security and workforce integrity. Employing risk-based assessments and proportional safeguards for CSP employees will ensure the integrity of AI systems while simultaneously fostering innovation and competition in the market.

III. Protecting American Competitiveness Abroad

In the rapidly developing landscape of artificial intelligence, it is imperative that the United States adopt a strategic approach to export controls and trade policy. A well-crafted trade policy will allow U.S. companies to access and compete in foreign markets, benefiting American economic and geopolitical interests while restricting adversary access to critical technologies.

A. Maintaining and refining export control policies

The U.S. must ensure that its export control policies are robust, adaptive, and effective in preventing adversaries from accessing critical AI technologies, while balancing this with the need to support U.S. companies in their ability to innovate and compete in foreign markets. These policies should be continuously updated to reflect technological advancements and applied consistently across all relevant sectors.

One critical need is to modernize the Bureau of Industry and Security. The Bureau needs to be provided with sufficient resources, potentially including utilizing AI systems, to successfully process licenses, monitor the supply chain and counter attempts to violate export controls. At the same time, a careful implementation of such modernization is needed to avoid creating undue burdens on U.S. businesses. In particular, the processes for obtaining export licenses require streamlining to limit regulatory burden on U.S. exporters. A risk-based approach may be useful in this streamlining, calibrating the level of scrutiny on a given export license request to the level of risk export of the product creates.

B. Limit foreign overregulation and non-tariff trade barriers

The U.S. must actively engage with foreign governments to oppose excessive restrictions on AI development and innovation. U.S. firms, representing leading capabilities in foundational models, basic research, advanced computing, and end-use tools, are at significant risk of discriminatory treatment. This includes challenging requirements that hinder innovation and U.S. exports. Such requirements can include excessive pre-deployment testing of AI models, restrictions on cross-border data flows, data localization requirements, forced disclosure of source code and algorithms, unbalanced copyright policy, country-specific technical standards that stifle model deployment, and discriminatory treatment of service suppliers.

For example, restricting the ability to transfer data between countries restricts the ability of companies to develop and deploy AI services globally. This harms the ability of companies that power the most prominent AI models from creating technologies that work in and reach new markets, while also undermining the ability of smaller businesses and startups that rely on these companies to build innovative applications and services on top of those AI models to grow to new markets.[7] Fostering international dialogue to ensure that regulations are not overly burdensome is essential to American international competitiveness.

Further, AI is increasingly integrated and embedded into a broader set of U.S. digital services in areas such as online search; recommendation algorithms for social media, online streaming, and e-commerce suppliers; and customer service, marketing, and translation, enhancements, to name just a few. The overall state of foreign trade barriers blocking companies' ability to introduce new AI services, a trend that is already visible in the European Union,[8] is becoming more prevalent as AI is built into these other legacy online services.

Most urgently, engagement with allies such as the EU and South Korea is crucial to shaping beneficial AI policies globally. Via collaborative engagement on issues like the EU's AI Act, proposed revisions to their Digital Markets Act and/or Digital Services Act, and Korea's Basic AI Law, the U.S. can help prevent regulations that unduly harm American businesses and establish unified international technical standards that promote innovation and the competitiveness of American industry abroad.

Longer-term, engagement with foreign partners to reach strong, enforceable commitments that promote the cross-border trade of services reliant on AI would ensure U.S. companies can develop and export digital services and products reliably to key markets.

Finally, foreign reciprocity in market access is essential. For example, if U.S. companies face restrictions on their access to Chinese markets, similar barriers should not be lifted to provide access to U.S. markets to Chinese AI companies. While absolute parity may be impossible, fair market access must be available at the very least.

C. Ensure that restrictions on export are transparent

Any designations that may affect or restrict the sale of AI technology to foreign countries or entities must be guided by transparent criteria, with clear timelines for when enforcement will begin and clear requirements for compliance. This transparency is essential for businesses to be able to successfully meet their compliance obligations, safeguarding the national interest while ensuring fairness and predictability. In particular, it is essential that any modifications to export controls be preceded by significant public and industry consultation, as well as appropriate Congressional oversight. Input from industry can ensure that export controls are not overly burdensome while still effectively providing export controls that avoid the provision of critical technologies to adversaries.

IV. Ensuring Uniform National AI Policy

AI presents huge potential to transform the national economy. State legislatures across the country are currently developing a wide array of AI-related legislation. While there may be unique local concerns in some areas, the vast majority of AI policy would be best served by a

[7] https://ccianet.org/wp-content/uploads/2024/12/CCIA-Comments-on-AI-Competitiveness-for-the-Presidents-Export-Council.pdf.
[8] https://www.cnbc.com/2024/06/21/apple-ai-europe-dma-macos.html; https://www.theverge.com/2024/7/18/24201041/meta-multimodal-llama-ai-model-launch-eu-regulations; https://www.euronews.com/next/2024/12/10/openai-releases-ai-video-creator-sora-but-it-wont-be-coming-to-europe-yet.

unified national policy that provides clear determination of the responsibilities of the various actors in the AI value chain. This may include sector-specific regulation, as appropriate.

A. State laws regulating AI should be preempted where federal laws exist

Generally, federal legislation should preempt state laws concerning the development and deployment of frontier AI models. This preemption should fully preempt state law in these areas, ensuring a unified federal approach. Such an approach ensures consistency and facilitates compliance for businesses operating across multiple states, preventing a fragmented regulatory landscape that could harm consumers, hinder innovation, and create unnecessary obstacles for industry growth.

However, while full preemption is generally preferable, states will still have a role to play. In particular, state legislatures are ideally suited to address specific gaps or unique local concerns that may not be not adequately covered by federal regulations. By starting from a presumption of preemption and permitting states to operate to fill gaps where necessary, this approach will ensure national uniformity in managing frontier AI while accommodating state-specific needs.

B. Federal agencies should clarify the application of existing law to AI

Federal agencies are encouraged to clarify how existing laws apply to AI technologies. As CCIA has noted repeatedly in the past, most issues with AI can be handled by existing law.[9] If AI is abused to harm consumers, it does not require a specific AI consumer protection law—existing laws such as the FTC's Section 5 authorities are sufficient. And if a harm is not currently illegal, the correct approach is to legislate to render the harm illegal regardless of who performs the action. Only where there is a risk or harm that is unique to AI systems and not present in the human or non-AI software equivalent is legislation required.[10]

In order to emphasize this point, federal agencies should review the laws they implement or enforce and provide guidance to stakeholders on how existing law applies to AI. Agencies should conduct thorough assessments to determine if existing laws sufficiently address risks associated with AI and consider targeted updates or interpretations as necessary. By doing so, we can avoid the creation of duplicative, inconsistent, and cumbersome new regulations. Leveraging current legal frameworks ensures efficient oversight without stifling innovation. This approach not only streamlines regulatory processes but also provides clarity for developers and deployers, facilitating the compliance necessary to protect consumers and fostering a predictable environment for AI development.

C. A sectoral approach should be employed where appropriate

As discussed above, many aspects of AI regulation are already addressed by existing law. And many other aspects of AI regulation can be applied evenly across all industrial sectors. However, even though many aspects of AI are common across all sectors, in some areas a sectoral approach might be needed. For example, sectoral regulations in specific industries where AI applications pose unique risks and requirements is essential. Industries such as healthcare, finance, and transportation have unique risks and requirements that require sectoral regulation.

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[9] *See, e.g.*, CCIA, *Understanding AI: A Guide to Sensible Governance* (June 26, 2023), https://ccianet.org/library/understanding-ai-guide-to-sensible-governance/.
[10] *Id.* at 2.

At the same time, those sectoral regulations may be unnecessary or actively harmful in other sectors, imposing a need to limit their applicability outside of the appropriate sector.

To support this strategy, the establishment of a center of technical expertise is proposed. This center would provide specialized knowledge and resources for each sector, ensuring that regulations are effective without imposing unnecessary burdens. By working closely with relevant agencies, this center can facilitate tailored oversight that addresses the distinct needs of different industries, promoting both safety and innovation. CCIA has previously suggested, and continues to suggest, that such a center should be housed within the National Institute of Standards and Technology (NIST).[11] NIST has significant technical expertise and the experience working across many sectors that such a center requires. They have also already developed significant expertise in AI through programs such as the NIST AI Risk Management Framework. At the same time, while they have extensive experience in coordinating technological standards, they may lack sector-specific experience and such a center may require new budgetary resources.

D. Clarifying the roles and responsibilities of various actors in the AI value chain

In discussing AI, it is critical to carefully define the various participants in the value chain and their associated responsibilities. The developer that trains a model is not necessarily the same entity that operates and deploys it. The developer might also not be the entity that assembles the training data; in some circumstances, a third party will be the entity that does that. The operator of an AI system may not be the deployer, but may instead be a third-party compute provider who is acting as a service provider to the deployer. And in some circumstances, responsibility will lie with the end user.

Given this complex value chain, it is essential to clearly define the roles and responsibilities of all actors in the AI value chain, assigning them to the actor best placed to address the concern. The developer of an AI technology will often not be the entity best suited to limiting abuses at the end-user level. Instead, regulation should ensure that developers of AI technology can focus on creating models, while deployers handle integrating these models into applications. Operators may focus on providing efficient and accessible systems for deployers to take advantage of, while dataset assembly entities may focus on providing datasets that provide high-quality data, including sets that may be tuned to particular needs. Establishing standards for transparency, liability, and compliance at each stage ensures accountability and addresses potential overlaps or ambiguities between the various actors. This clarity prevents disputes and gaps in accountability, ensuring that responsibilities are well-defined and understood across the AI ecosystem. It also reduces the potential for inconsistent legislation and conflicting obligations on a worldwide scale; legal frameworks in other nations have recognized the value in clearly defining roles and have done so.

V. Maintaining Necessary Data Access for AI Development

One critical input to artificial intelligence development is data access. Especially important to a robust data access policy are balanced copyright rules, reasonable transparency measures, and a regulated ability to use personal data to deliver services that are needed and wanted by consumers and businesses alike. These types of data access policies are essential components in driving AI success.

[11] *Id.* at 8.

A. Balanced copyright policy is essential to AI success

While copyright can provide a reward for creativity, an overbroad conception of copyright and its over-enforcement is fatal to creativity. Balanced copyright laws, including fair use[12] and text-and-data mining (TDM) exceptions, provide a legal framework that supports innovation, free speech, the public interest, and the rights of content creators alike.[13] Because, under the Copyright Act, all works of authorship are copyrighted at the moment of fixation,[14] it is crucial to provide the safety valves of fair use and TDM exceptions in order to enable learning and communication.[15] This is no less true for AI than for human learners. These rules enable AI systems to learn from existing knowledge, thereby accelerating scientific discoveries and societal progress. For instance, AI applications can sift through vast libraries of scientific papers to identify patterns or generate hypotheses, as seen in drug discovery efforts where AI analyzes thousands of studies alongside background data to find potential candidate compounds.[16]

Even assuming that AI training on copyrighted data is the type of use that qualifies as covered by the limited set of rights copyright provides, a question which remains unsettled, CCIA believes that it is still a fair use under existing caselaw to engage in such training. A wide range of cases currently making their way through the courts will establish the rules and boundaries for AI training. However, given that this issue is likely to remain unsettled for some time as it is appealed through the courts, it could be beneficial for the Administration to work with Congress to clearly establish that using data for AI training is a fair use via legislation.

Availability of data access via fair use also limits barriers to entry by removing the requirement to negotiate with data holders during model development and experimentation. This is not to say that data holders have no ability to extract value from their data, even if training on it is a fair use. For example, a range of data licensing deals have been signed by AI companies. While these deals include a copyright license component, the key value add of data holders is in the provision of data in a high-quality structured fashion. Rather than being forced to scrape the Web or OCR copies of newspapers or books, negotiation allows AI developers to obtain the needed data in a more useful form. This benefits both the AI developer and the data holder, who receives compensation for their work in providing the data. A strong fair use regime thus protects artists, rightsholders, and AI innovators by restraining the overuse and abuse of copyright to block new art and innovation alike.

The Administration can best support AI by actively advocating for these exceptions, both at home and abroad in the form of fair use or TDM exceptions. This advocacy can include regulatory policy work, direct engagement in foreign jurisdictions, and amicus support in the courts.

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[12] 17 U.S.C. § 107.

[13] *See, e.g.*, CCIA, *Fair Use in the U.S. Economy: 2017* (June 9, 2017), https://ccianet.org/research/reports/fair-use-in-the-u-s-economy-2017-edition/.

[14] 17 U.S.C. § 102.

[15] *See, e.g.*, *Stewart v. Abend*, 495 U.S. 207 (1990) (fair use "permits courts to avoid rigid application of the copyright statute when, on occasion, it would stifle the very creativity which that law is designed to foster.")

[16] *See, e.g.*, Derek Lowe, *Snake Antivenoms, Computed*, In The Pipeline (Jan. 16, 2025), https://www.science.org/content/blog-post/snake-antivenoms-computed.

B. Transparency to build trust

Appropriate transparency is crucial for fostering trust in AI technologies. Mechanisms like watermarking synthetic content help users discern between real and generated material, enhancing accountability. However, transparency must be implemented thoughtfully. Standards should avoid being overly burdensome, protecting free speech and trade secrets while ensuring usability. This balance is vital to encourage innovation without compromising security or functionality.

While watermarking techniques can usefully address some aspects of transparency, there are other concerns it is less suited to address, especially regarding text generation where watermarking is more difficult or even impossible. To accommodate some of these other concerns, other approaches are needed. For example, appropriate documentation of training practices and outcomes can help enhance trust and transparency. To the extent disclosure requirements are imposed, they should remain at a relatively high level. A high-level requirement allows individual companies with better understanding of their company's specific technology to tailor their disclosure to balance between interests such as avoiding the disclosure of competitive information like the specific contents of a training data set and providing sufficient information for the public to understand what general forms of information were used and how that information was used.

Transparency requirements may also benefit from a sector-specific regulatory approach. Safety-critical applications, such as automotive or medical AI uses, may require a higher level of transparency and documentation in order to build public trust. Regardless of the sector, however, it is critical that collaboration among stakeholders—AI developers, policymakers, ethicists, and users—be a part of the development process. Existing attempts at transparency regulation often suffer from a lack of understanding of the technology and the law alike, such as laws that propose to require documentation of the presence and owner of every piece of copyrighted material used in training. Since, as noted above, every piece of information is copyrighted at the time of fixation, this amounts to a requirement to document the copyright owner of every single piece of content scraped from the public Internet, an impossible task.

By taking advantage of the knowledge various stakeholders can bring to the table, a balanced regulatory approach to transparency will protect the rapid pace of innovation in AI that American ingenuity has delivered while enhancing public trust in the technology.

C. Personal data and model training

AI models, particularly large language models (LLMs), rely heavily on diverse datasets that often include personal information. This data is essential for understanding context and nuances in language usage by different types of entities ranging from individuals to small businesses to large corporations, enabling tasks such as sentiment analysis. For example, correctly referencing names in historical events ensures accurate and detailed outputs, which would be hindered without access to such data.

While some types of personal data can present privacy concerns, as addressed below, removing or masking personal data from training datasets can impair a model's ability to understand language effectively and deliver services the public expects. This underscores the need for balanced approaches that respect privacy while maintaining model quality. Retraining models to exclude such data is resource-intensive and impractical on short timelines, highlighting the necessity of flexible solutions.

VI. Avoiding Foreign Abuse of Patents Against AI

While significant attention has been given to the implications of copyright for AI competitiveness, less attention has been given to the problem of patent abuse against American AI companies. As a rapidly developing area of technology, patents can be extremely important to AI technology. They can also prove fatal to startups when abused.[17] More than half of all U.S. patents issue to foreign entities, including China, and many of those patents cover AI technology. Some of these patents contain true innovation, but many are likely to prove invalid if asserted. One Chinese patent expert even characterized 90% of Chinese patents as "trash."[18] However, proving invalidity in district court cases is an incredibly expensive proposition, one which can take years, meaning that even those "trash" patents can be weaponized against American AI companies. Even worse, during district court litigation the foreign entity can engage in discovery to obtain sensitive technical information from the American innovator.

This is particularly concerning given the history of Chinese entities turning to patent litigation when excluded from the U.S. market. This is not an imaginary concern. After Huawei was excluded from the U.S. market in 2019, it turned to patent litigation.[19] Huawei filed lawsuits against American companies like Verizon[20] seeking $1 billion in damages.[21] Ultimately, Verizon settled under undisclosed terms. The Huawei assertion campaign is a concerning model for what is likely to happen to U.S. companies if and when foreign adversary companies with AI patents begin assertion campaigns. Beyond direct assertion, there is significant abuse of the U.S. patent system by undisclosed foreign funders—many believed to be Chinese—who pay for third parties to assert patents against American innovators with the aim of reaping a windfall benefit in damages.[22]

There are two primary defenses against this type of abuse. The first is the Patent Trial and Appeal Board (PTAB). This component of the U.S. Patent and Trademark Office is staffed by technically and legally trained administrative patent judges (APJs), who review the validity of patents when challenged. These challenges are an order of magnitude less expensive than district court litigation; they are also at least as, if not more, accurate in determining invalidity than lay judges and juries without the technical background of APJs. They also primarily benefit American companies, with nearly two thirds of petitions brought by U.S. companies.[23] In contrast, in any given year between 65% and 80% of patents challenged at the PTAB are owned by foreign entities.[24] The PTAB provides an effective forum for American companies, including AI companies, to defend themselves, and the Administration should reject efforts to weaken the

[17] https://www.kickstarter.com/projects/aiforeveryone/mycroft-mark-ii-the-open-voice-assistant/posts/3729060.
[18] https://www.cigionline.org/articles/what-do-chinas-high-patent-numbers-really-mean/.
[19] https://www.reuters.com/article/idUSKCN1SL2QX/.
[20] https://www.wsj.com/articles/huawei-settles-two-patent-lawsuits-it-filed-against-verizon-11626105146.
[21] https://www.techdirt.com/2019/06/17/huawei-now-using-patent-claims-to-demand-1-billion-verizon-as-us-tries-to-c
[22] https://issa.house.gov/media/press-releases/issa-introduces-legislation-reforming-third-party-financed-civil-litigation.
[23] https://www.unifiedpatents.com/insights/2019/12/11/new-study-suggests-that-us-companies-have-received-the-greatest-benefit-from-post-aia-proceedings.
[24] *Id.*

PTAB both by regulation and legislatively. It should also reject efforts to reimpose the *Fintiv* rule that restricts American companies' ability to access the PTAB.

The second defense is high-quality patent examination. U.S. patent examiners have a difficult job, having an average of only 19 hours to assess whether to grant a patent. This, in turn, leads to numerous low-quality patents issuing from the Patent Office. Providing examiners with more resources would help to ensure that they can detect low-quality patents that might be abused to harm American AI innovation before those patents issue, weeding out the patents likely to ultimately prove invalid and thereby increasing the value of granted patent rights.

The combination of an initial high-quality examination and the availability of review via the PTAB will help to ensure that innovation in artificial intelligence is rewarded, while preventing abuse of our patent system against those innovators.

VII. Open-Source AI Serves Critical Functions

CCIA members have actively developed both open-source and closed-source models. Other CCIA members have integrated open-source and/or closed-source models into their commercial operations. Both open- and closed-source approaches have benefits and disadvantages and both will be part of a well-functioning AI ecosystem. Industry experience with operating systems, in which closed-source operating systems such as Microsoft Windows function alongside open-source options such as Linux and mixed open/closed approaches such as Apple macOS, is illustrative here—each option serves distinct functions and they co-exist in the market to the advantage of every computer user.[25]

Artificial intelligence is no different. Open-source models can help to increase competition, including by bringing down the cost of experimenting with AI technology for small and medium enterprises. Closed-source models can provide benefits such as enhanced support and the availability of simpler scaling options. Prior work, including a detailed report from NTIA,[26] has outlined a number of other benefits and risks for open-source AI, including benefiting cybersecurity defense by enhancing attack detection and mitigation,[27] research and development through wider access and fine-tuning of models to particular research agendas,[28] and avoiding AI monocultures.[29] CCIA supports NTIA's recommendation of continuing to gather information regarding the risks and benefits of open-source models.[30]

Beyond concrete benefits such as those described above, open-source AI also helps to ensure that American values are enshrined in the global AI ecosystem. If the U.S. were to restrict open-source models, it would not deter adversaries from developing their own models. It would be highly beneficial to the free world if neutral foreign nations who seek to adopt artificial

[25] While Windows is characterized here as closed source, it incorporates significant open-source components. Similarly, although Linux itself is open-source, a number of closed-source programs can be integrated into it to provide enhanced functionality, particularly with respect to hardware support.
[26] NTIA, *Dual-Use Foundation Models with Widely Available Model Weights* (July 2024), https://www.ntia.gov/sites/default/files/publications/ntia-ai-open-model-report.pdf.
[27] *Id.* at 17.
[28] *Id.* at 28.
[29] *Id.* at 32.
[30] *Id.* at 3.

intelligence technologies adopt models that reflect democratic values rather than being forced to adopt models that inherit the values of adversary nations.[31]

Neither open-source nor closed-source models can, standing alone, fulfill the transformative promise of artificial intelligence. Neither, standing alone, can mitigate all potential AI risks. It is critical that any regulatory actions maintain the ability to develop both open- and closed-source models.

VIII. Protecting Privacy Without Blocking Development

Much like transparency, a balanced and effective privacy regime is key to building AI trust. But a single privacy regime that applies to all parties in the AI ecosystem without recognizing the disparate roles that entities may play is likely to prove both ineffective in protecting privacy and in ensuring AI innovation. Generative AI technologies operate in two distinct phases—the development and training of the AI model and the building and operation of applications that use the AI model. Only the building and operation of applications is directly user-facing. These two phases have distinct uses for data, privacy implications, risk levels, and opportunities to create and implement safeguards.

Generally, privacy controls will be more effective on the application or output side. This side carries both a greater risk of harm—for example, personal data disclosure—and a greater ability to implement safeguards. Personal data leakage or hallucinations about a non-public living person can often happen through interaction with a product but rarely happen during model development and training. Detecting and correcting such harms is also significantly more practical at the product or output stage than during training.

In particular, the application layer provides opportunities to protect against inappropriate, offensive, harmful, or otherwise privacy-destroying content. This can range from legal mechanisms, such as usage policies and enforcement against those who violate them, to technological mechanisms that provide limits on how the AI application interacts with personal data. Fine-tuning can also help to reduce the incidence of improper outputs. Finally, screening mechanisms that examine the output of the AI application, such as filters and classifiers, can help to avoid improper output. These mechanisms may themselves be AI applications.

In developing AI privacy regulation, it is critical for the administration to bear in mind the differences between the two major phases of generative AI development. Applying a single universal regime to both phases is likely to both fail to protect the privacy of American citizens and to harm the development of AI technology in the United States.

IX. Conclusion

CCIA appreciates the opportunity to comment on these critical issues and would be happy to assist the NSF and OSTP with any further requests for information.

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[31] *Id.* at 23.

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